Dr. Magdalena Oryaëlle CHEVREL

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Laboratoire Magmas et Volcans
Université Clermont Auvergne
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6 av. Blaise Pascal
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ACADEMIC POSITIONS

Since 2019 Researcher at the Institut de Recherche pour le Développement IRD *Laboratoire Magmas et Volcans (LMV), Université Clermont Auvergne (UCA), France*

2022-2025 Hosted researcher at the Observatoire Volcanologique du Piton de la Fournaise - Institut de Physique du Globe de Paris (OVPF-IPGP)*, La Réunion, France*

Since 2022 Volunteer assistant professor, Earth Sciences Faculty of University at Buffalo, *USA*

2018 – 2019 Postdoc Researcher ANR funding, *LMV, UCA, France*

2016 – 2017 Postdoc Fellowship ClerVolc, *LMV, UCA, France*

2014 – 2015 Postdoc Fellowship DGAPA *Universidad Nacional Autónoma de México (UNAM), Mexico*

EDUCATION AND QUALIFICATIONS

2024 HDR : Habilitation à diriger des recherches (French degree to lead research), *Université Clermont Auvergne (UCA), France*

2013 PhD : Physical and Experimental Volcanology. "Rheology of Martian Lava Flows: An Experimental Approach". *Ludwig Maximilians Universität (LMU), Germany*.
Supervision: Don Dingwell, David Baratoux
Defended on May 8, 2013, awarded: "*summa cum laude*".

2009 Master in Earth and Planetary Sciences *Université Toulouse III, France*

2008 Trans-Atlantic science student exchange program *University of Calgary, Canada*

2007 Licence in Earth and Planetary Sciences *Université Toulouse III, France*

July 2008 Research internship *Institute for Study of the Earth's Interior (ISEI), Japan*

April 2008 Research internship *University of Calgary, Canada*

OBSERVATORY ACTIVITIES

Since 2018: Monitoring, modeling and anticipation of lava flows during eruptions at Piton de la Fournaise OVPF-IPGP, France

April 2019 – Feb. 2020 and from Oct. 2022 to 2025: Participation in weekly on-call duties and reinforced on-call duties during eruptions (processing of seismic data, monitoring of activity). Field surveys and support to OVPF teams for field missions during and outside eruptions (lava sampling, GPS surveys).

FUNDINGS

Scientific projects

2026 « Measuring the lava viscosity in the Field » PI 100% 100 000€, ANR Templin France

2026 JEAI VIVOMAC « Vivre au milieu des volcans au Mexique et en Amerique centrale: Risques et Patrimoine Naturels » Co_PI 50% - IRD Correspondent, 50 000€, IRD France

2024 « Tackling Multiphase Lava Rheology at its Origins» Co-PI 50 % (PI: Stephan Kolzenburg). EAR #2420723, $410,626.00; National Science Foundation, UB, USA

2023 « Measuring the viscosity of lava using the new portable field viscometer » Principal Investigator (PI) 100%. 10 000€ Clervolc, LMV, UCA, France

2023 « Rheology for near real time forecasting of lava flows » Collaborator (PI: Stephan Kolzenburg). EAR 2223098, 2/01/2023-1/31/2026, $410,626.00; National Science Foundation, UB, USA

2022 « RAPID: Deployment of a Field Rheometer Prototype » Collaborator (PI: Stephan Kolzenburg) EAR 2241489, 8/15/2022-7/31/2023, $52,582 National Science Foundation, UB, USA
2021 « From magma ascent to deep-seated-ocean lava flow emplacement: the case study of the ongoing (since 2018) submarine eruption offshore Mayotte » PI 40% (co-PI Lucia Gurioli and Etienne Medard). 125 000€ Clervolc, LMV, UCA, France
2020 « A new portable field viscometer » PI 100%. 10 000€ Cheque Recherche Innovation, Cap20-25, Isite, UCA France
2020 « Rheology of the lava flows of Sierra Negra and Cerro Azul, Galapagos, Ecuador ». PI 100%. 4 000€ ClerVolc, LMV, UCA, France
2017 « Building a new portable field viscometer ». PI 100%. 4 000€ Actions Incitatives of the Observatoire de Physique du Globe de Clermont-Ferrand, UCA, France
2016 « The thermo-rheological evolution of lava flows » PI 100%. Two years salary + 10 000€, Auvergne Fellowship et ClerVolc, 2016 Auvergne Fellowship, ClerVolc, LMV, UCA, France
2014 «Eruptive activity and physical properties of magmas feeding enigmatic shield volcanoes in the Michoacán-Guanajuato Volcanic Field: El Metate case study ». PI 100%. Two years salary. DGAPA, UNAM, Mexico

Long term missions, mobilities and assignments abroad and overseas IRD
2022 - 2025 Assignment to OVPF-IPGP, La Réunion, France - 3 years
2020 Long-term mission at OVPF-IPGP, La Réunion, France - 3 months
2021 Research mobility IG-Ecuador to LMV UCA, France - 2 months

Student grants
2013 IAVCEI, Kahoshima, Japan
2012 IAVCEI, Cities on Volcanoes 7, Colima, Mexico
2012 European Space Agency to participate at EGU, Vienna, Austria
2010 AGU, 2010 fall meeting, San Francisco, USA
2009 PhD grant (3 years) from the International Graduate School, Network of Bavaria, LMU, Germany
2008 International Student Intern Program at Misasa, Institute for Planetary Materials (IPM), Okayama University (Misasa, Tottori), Japan
2007 Trans-Atlantic Science Student Exchange Program. University of Calgary, Canada

AWARDS
2025 : Hors classe
2019, 2025 Prime d'encadrement doctorale et de recherche
2018 Speed poster award during the Workshop Interdisciplinaire sur la Sécurité Globale, *Paris, France* 2013 PhD defense awarded with “Summa cum laude”, *Munich, Germany*
2011 Poster award, 9th Silicate Melt Workshop, *La Petite Pierre, France*

LANGUAGE SKILLS
French (mother tongue), English (level C1), Spanish (level C1)

PHD CANDIDATE SUPERVISION
2023 – 2025 Martin Harris « Mesuring lava viscosity in the field. » Co-supervision at 50%, with Stephan Kolzenburg *Université at Buffalo, USA*
2021 – 2024 Pauline Verdurme “Emplacement dynamics of the submarine eruption offshore Mayotte”. Co-supervision at 40% with Lucia Gurioli et Etienne Médard *LMV, UCA, France LMV, UCA, France*

PhD JURY MEMBER
2025 : Reviewer: François Lotter “The Eruption Type Variability of Basaltic Shield Volcanoes: The Example of the Karthala and Piton de la Fournaise” LGSR Univ. Réunion
2021-2024 Reviewer. Jacob Brauner “Volcanic Hazards on Bioko (Equatorial Guinea) – Remote Sensing, Mapping and Modelling” *Drexel University, USA*
2022 Substitute member. Francesco Massimetti “Thermal remote sensing of volcanic activity by using Sentinel-2 and Landsat-8: an improvement of the MIROVA system” *University of Turin, Italy*
2021 Examiner. Jean-Marie Prival “On the emplacement dynamics of highly-viscous, silicic lava flows” *LMV, UCA, France*
2017 Invited member. Silvia Vallejo Numerical models of volcanic flows for an estimation and delimitation of volcanic hazards, the case of El Reventador volcano (Ecuador). *LMV, UCA, France*

MASTER PROJECT SUPERVISION

2026 E. Rampa « Etude retrospective des proprieties thermo-rhéologique des coulees de lave du Piton de la Fournaise ». LMV, UCA France. Encadrement 100%

2025 C. Petit « Modélisation des grandes coulées de lave hors- enclos au Piton de la Fournaise ». OVPF-IPGP-LGSR France. Co-encadrement 50%

2024 - 2025 J. Diamico « Automated Near-Real Time Rheological Flow Modeling for Nyiragongo Volcano (D. R. Congo) » Co-supervision 50% with Stephan Kolzenburg. UB, USA.

2022 T. Lemaire. « Using Volcflow as an operational tool for modelling lava flows at Piton de la Fournaise. » Co-supervision 50% with Karim Kelfoun. Year 2, LMV, UCA, France.

2022 L. Maingault. « Test du modèle thermo-rhéologique PyFLOWGO sur les coulées de lave de l'éruption de La Palma, 2021. » Co-supervision 50% with Andrew Harris, Year 1, LMV, UCA. France.

2022 S. Gueho. « Analyse texturale d'échantillons provenant de l'éruption de 2021 du Cumbre Vieja, La Palma. » Co-supervision 50% with Andrew Harris and Lucia Gurioli. Year 1, LMV, UCA. France.

2020 J. Fort. « Modélisation des coulées de lave au Piton de la Fournaise avec VolcFlow. » Co-supervision 50% with Karim Kelfoun. Year 2, LMV, UCA, France.

2019 A. Pawlak. « Remonté de péridotites par des laves basanitiques. » Co-supervision 50% with Didier Laporte. Year 1, UCA, France.

2018 – 2019 N. Reyes-Guzman. « Emplazamiento de los flujos de lava de la región occidental de la cuenca lacustre de Zacapu, Michoacán. » Co-supervision 50% with Claus Siebe. UNAM, Mexico.

2018 – 2019 I. Ramírez-Uribe. « Emplazamiento de los flujos de Rancho Seco y estructuras volcánicas vecinas (Michoacán, México). » Co-supervision 50% with Claus Siebe. UNAM, Mexico.

2018 P. Thao and J. Fort. « Le rôle de la viscosité dans les modèles de coulées de lave. » Co-supervision 50% with Andrew Harris. Year 1, LMV, UCA. France.

2017 A. Ajas. « L'influence des arbres sur l'avancée d'une coulée de lave, example au Kilauea, Hawaii. » Co-supervision 40% with Andrew Harris and Lucia Gurioli. Year 2, LMV, UCA, France.

2014 – 2016 JR de la Fuente. « Geología volcánica (radiometric dating, geochemistry) del area de Paracho-Cheran, Michoacán, Mexico. » Co-supervision 50% with Claus Siebe. UNAM, Mexico.

2011 – 2012 L. Debiasi. « The complex rheology of crystallizing melts. » Co-supervision 80% with Corrado Cimarelli. LMU, Germany.

UNDERGRADUTATE PROJECT SUPERVISION

2023 R. Portigliatti-Presa. « Etude de la modélisation et du comportement visqueux d'une coulée de lave. » External supervision. Ecole des Mines-Telecom de l'Université de Lille

2022 E. Pasci. « Determination of morphometric parameters and simulation of lava flows in the Chachani Volcanic Complex – Arequipa. » External advisor. Supervision by Ben Van Wyk de Vries (LMV) and Nelida Manrique INGEMMET, Pérou

2022 E. Rodriguez. « Cartografía y análisis de la eruption 2018 del volcán Fernandina, Galápagos, Ecuador. » Co-supervision 30% with Benjamin Bernard IG-EPN, Ecuador.

2020 – 2021 H. Calderon. « Cartografía y análisis de la eruption 2018 del volcán Sierra Negra, Galápagos, Ecuador. » IG-EPN, Equateur. Co-supervision 30% with B. Bernard and S.Vallejo IG-EPN, Ecuador.

2015 – 2016 N. Reyes-Guzman. Geología volcánica de la región occidental de la cuenca lacustre de Zacapu, Michoacán y su importancia para la arqueología. *UNAM, Mexico*

TEACHING

Since 2014 Lectures on Effusive volcanism (2h every year) in the General Volcanology Course, Master Level, *UNAM, Mexico*

2022, 2023 "Le métier de volcanologue" Licence 3*, UCA, France*

2022 "Haroun Tazieff, Les Scientifiques MADE IN France ", École Doctoral des Sciences Fondamentales*, UCA, France*

2016 – 2017 Rock characterization and petrology of igneous rocks (15 h), first year university level, *UCA, France*

2014 – 2015 Lectures of volcanology and geodynamics (22 h) for undergraduate and graduate students at the Science faculty of the *UNAM, Mexico*

2013 Lecture and exercise on lava rheology measurements for the NEMOH Network School (one day). *LMU, Germany*

2010 – 2012 In charge of the presentation of the high-temperature and rheological investigations lab for the international short course "Melts, Glasses and Magmas" *LMU, Germany*

INVOLVEMENT WITH THE COMMUNITY

2026 – current: Elected member of the Section 20 of Conseil National de la Recherche Scientifique (CNRS)

2024-2025 Elected member of the Scientific Committee of IRD
Since 2022 Member of the Scientific Committee of Observatoire Science de l'Univers - Réunion
2021-2022 Member of the Educational Advice Committee of Ecole de l'Observatoire Physique du Globe de Clermont
Since 2021 Member of the steering committee of the project MARMOR (Marine Advanced geophysical Research equipment and Mayotte multidisciplinary Observatory for research and Response)
Since 2010 Member of the International Association of Volcanology and Chemistry of the Earth's Interior, Collaborative volcano research and risk mitigation, European Geosciences Union et American Geophysical Union
2020-2022 In charge of seminar organization at *LMV, UCA, France*
2015-2016 In charge of seminar organization at *UNAM, Mexico*
2017-2020 Representative member of the Early Career Researcher Network of IAVCEI
2013-2020 Active member of the Early Career Researcher Network of IAVCEI
2005-2009 Student association for Geology (MAGMA) *University Toulouse III, France*

REVIEWS AND EDITORIAL INVOLVEMENT

Since 2026 Nominated Deputy Editor of VOLCANICA
Since 2018 Elected member of the editorial committee of the first open access journal in volcanology: VOLCANICA
2020-2021 Editor in charge of the special issue collection of Reports from each of the official volcano monitoring agencies in Latin America. VOLCANICA
Since 2010 Multiples reviews for international journals (Geology, Bulletin of Volcanology, Geophysical Research Letters, Journal of Geophysical Research, etc..)

INVITED SEMINARS

2025 Breaking barriers in science: Insights from Latin America's Volcano Observatories dual-language Special Issue IAVCEI, Genève, Suisse
2023 Monitoring and modelling lava at Piton de la Fournaise, *USGS*.
2023 Tracking and anticipating lava flow emplacement, Application to Piton de la Fournaise, *UNAM, Mexico.*
2023 Measuring the viscosity of lava in the field using a portable rotational rheometer. IAVCEI General Assembly, Rotorua, NZ
2022 Suivi et modélisation des coulées de lave. 1ère conférence internationale sur la gestion des volcans des Virunga, *Goma, RDC*
2022 Los volcanes y su monitoreo. *Ministerio de Energia y Minas, Cuba*
2021 Tracking and anticipating lava flow emplacement, Application to Piton de la Fournaise, *Univeristy of Buffalo, USA* & *Univeristy of Michigan, USA*
2021 Anticiper et surveiller les coulées de lave : application au Piton de la Fournaise, *UCA, France*
2020 Anticiper et surveiller les coulées de lave : application au Piton de la Fournaise, *Univeristé de la Réunion, France*
2020 La reologia de los flujos de lava : medidas, observaciones y modelos, Instituto de Geofisica, *Escula Polytechnica, Ecuador*
2018 L'activité Récente au Kīlauea, Hawaii, Les Mercredis de la Science, *Clermont-Ferrand, France*
2018 Constraining thermo-rheological properties of lava : from lab and field experiments to applications for modelling, *Cities on Volcanoes, Naples, Italy*
2017 Rheological control of lava flows emplacement, British Society of Rheology Mid-Winter Meeting, *University of Bristol, UK*
2015 La erupción del volcán escudo El Metate AD 1250 (Michoacán): La erupción mas voluminosa durante el Holoceno en México. *Centro de Geosciencias, Quérétaro, México*
2013 Volcanism on Mars: The rheology of lava flows, *Universidad de Bogota, Colombia*

H-INDEX AND CITATIONS

h-index (June 2026): 24 with 1397 citations (Scopus) and 25 with 1867 citations (Google Scholar)
> 70 international meeting proceedings, including > 15 oral presentations

LIST OF PUBLICATIONS

57. Guilbaud M-N, Mérour E, van Wyk de Vries B, Ortega-Larrocea M P, Cram S, Shires C, **Chevrel O,** Martínez-Báez Téllez M F, Zaragoza Alvarez S E, Morgan-Proux C. (2026) From ecological sensemaking to sensegiving: A way to develop a scientific project that reaches out to society. Front. Earth Sci. 14:1694641. https://doi.org/10.3389/feart.2026.1694641
56. Peltier A, Villeneuve N, Boissier P, Brunet C, Canjamalé K, Catherine P, **Chevrel MO**, Derrien A, Di Muro A, Desfete N, Duputel Z, Fontaine F, Frangieh M, Garavaglia L, Griot C, Journeau C, Kowalski P, Laborde C, Lauret F, Pesqueira

F, Richter N, Smittarello D, Vaitilingom A. (2026) Ten years of GNSS field campaigns covering a full eruptive cycle at Piton de la Fournaise (2014-2023). Bull Volcanol 88, 19 (2026). https://doi.org/10.1007/s00445-026-01944-2; https://theses.hal.science/FST-REUNION/hal-05499401v1

55. Harris M, Kolzenburg S. **Chevrel MO** (2025b) The Viscosity of Multiphase Lava: New Insights from Integrating Laboratory and Field Measurements. Earth Planet. Sci. Lett. 671 (2025) 119642. https://doi.org/10.1016/j.epsl.2025.119642 https://hal.science/hal-05285226/
54. Verdurme P, **Chevrel MO**, Médard E, Gurioli L, Berthod C, Brückel K, Komorowski JC and Bachèlery P. (2025) Lava rheological evolution and flow emplacement during the deep submarine Fani Maoré eruption (Mayotte) Journal of Geophysical Research: Solid Earth, 130, e2024JB030363. https://doi.org/10.1029/2024JB030363
53. Campus A, Villeneuve N, **Chevrel MO**, Peltier A, Di Muro A, Coppola D. (2025) Effusion rates trend analyses at Piton de la Fournaise (2000-2023): review of 24 years of space-based thermal observation. Journal of Geophysical Research: Solid Earth, 130, e2024JB030962. https://doi.org/10.1029/2024JB030962
52. **Chevrel O**, Harris A (2025). Monitoring Lava Flows. In: Spica, Z., Caudron, C. (eds) Modern Volcano Monitoring. Advances in Volcanology. Springer, Cham. https://doi.org/10.1007/978-3-031-86841-2_10. https://uca.hal.science/hal-04160974
51. Brauner J, **Chevrel MO**, Walter T R, Sealing CR, Vanderkluysen L (2025) Lava flow hazard assessment on Bioko Island (Equatorial Guinea) - A probabilistic approach using Q-LavHA. Volcanica, 8(1): 287–304. https://doi.org/10.30909/vol/wgfj4650
50*. Hamon C, Pereira G, Aubry L, **Chevrel MO**, Siebe C, Quezada O, Reyes-Guzmán N. Production of metates in central Mexico: techniques (know-how) and chaine operatoire of a traditional lithic craft in Turícuaro (Michoacán, Mexico). Journal of Archaeological Science: Reports. Vol. 64:105153 https://doi.org/10.1016/j.jasrep.2025.105153
49. Okumura S, Bain AA, **Chevrel MO**, Kendrick JE, Llewellin EW, Pistone M, Vona A and Whittington A. (2025) CHAPTER 2.3 "Physical properties of magmas and their evolution during storage, transport, eruption and emplacement" The Encyclopedia of Volcanoes, 3rd Edition Eds. Bonadona et al. https://doi.org/10.31223/X5ST63
48. Harris M, **Chevrel MO,** Parsons T, Latchimy T, Moreland WM, Thordarson T, Holskuldsson A, Clerc-Payet M, Kolzenburg S. (2024) Real-time, in-situ viscosity mapping of active lava. Geology. https://doi.org/10.1130/G52558.1; https://cnrs.hal.science/hal-04797214
47. Harris M, Kolzenburg S, I Sonder, **Chevrel O**. (2024) A new portable penetrometer for measuring the viscosity of active lava. *Review in Scientific Instrument. Vol. 95, 065103* https://doi.org/10.1063/5.0206776, https://hal.science/hal-04663298
46. Verdurme P, Gurioli L, **Chevrel MO**, Medard E, Berthod C, Komorowski JC, Harris A, Paquet F, Cathalot C, Feuillet N, Lebas E, Rinnert E, Donval JP, Thinon I, Deplus C and Bachèlery P. (2024) Magma ascent and lava flow field emplacement during the 2018–2021 Fani Maoré deep-submarine eruption insights from lava vesicle textures. *Earth and Planetary Science Letters. Vol: 636, 118720.* https://doi.org/10.1016/j.epsl.2024.118720 https://uca.hal.science/hal-04555016
45. Hamon C, Pereira G, Aubry L, **Chevrel MO**, Siebe C, Quezada O, Reyes-Guzmán N. (2024) Quarrying volcanic landscapes: territory and strategies of metate production in Turícuaro (Michoacan). Accepted 03/2024. *Geofísica Internacional*
44. Hamon C, Pereira G, **Chevrel MO**, Aubry L, Siebe C, Quesada O, Reyes N. Present use and production of « metates » in Turícuaro (Michoacán, Mexico): deciphering the evolution of food preparation practices. *Ethnoarchaeology.* *https://doi.org/10.1080/19442890.2023.2280379*
43. **Chevrel MO**, Latchimy T, Batier L, Delpoux R, Harris M, Kolzenburg S. A new portable field rotational rheometer for high-temperature melts. *Review in Scientific Instrument 94, 105116 (2023)* *https://doi.org/10.1063/5.0160247*
42. **Chevrel MO**, Villeneuve N, Grandin R, Froger JL, Coppola D, Massimetti F, Campus A, Hrysiewicz A, Peltier A (2023) Report on the lava flow daily monitoring of the 19 September - 05 October 2022 eruption at Piton de la Fournaise, *Volcanica: 6(2): 391–404.* *https://doi.org/10.30909/vol.06.02.391404*
41. Flynn ITW, **Chevrel MO**, Crown DA, Ramsey MSR. The Effects of Digital Elevation Model Resolution on the PyFLOWGO Thermorheological Lava Flow Model. *Environmental Modelling & Software, Vol. 167: 105768,* *https://doi.org/10.1016/j.envsoft.2023.105768.*
40. Flynn ITW, **Chevrel MO**, Ramsey MSR. (2023) Adaptation of a thermorheological lava flow model for Venus conditions. *Journal of Geophysical Research: Planets* Vol. 128, e2022JE007710. *https://doi.org/10.1029/2022JE007710*
39. Verdurme P, Lelosq C, **Chevrel MO**, Pannefieu S, Medard E, Berthod C, Komorowski JC, Bachèlery P., Neuville D., and Gurioli L. Viscosity of silicate melts from the active submarine volcanic chain of Mayotte. *Accepted 16/01/23- Chemical Geology*

38. Berthod C, Komorowski JC, Gurioli L, Medard E, Bachèlery P, Bession P, Verdurme P, **Chevrel MO**, Di Muro A, Peltier A, Devidal JL *et al.* (2022) Temporal magmatic evolution of the Mayotte offshore eruption revealed by in situ submarine sampling and petrological monitoring. *Comptes Rendus. Géoscience, Online first (2022), pp. 1-29. https://doi.org/10.5802/crgeos.155*
37. **Chevrel MO**, Harris A, Peltier A, Villeneuve N, Coppola D, Gouhier M, Drenne S. (2022) Volcanic crisis management supported by near real time lava flow hazard assessment at Piton de la Fournaise, *Volcanica* 5(2), pp. 313–334. https://doi.org/10.30909/vol.05.02.313334
36. Prival JM, Harris AJL, Zanella E, Test CR, Gurioli L., **Chevrel MO** and Biren J. Emplacement dynamics of a crystal rich, highly viscous trachyte flow of the Sancy stratovolcano (France) *GSA Bulletin 2022 https://doi.org/10.1130/B36415.1*
35. Smittarello D, Smets B, Barrière J, Michellier C, Oth A, Shreve T, Grandin, R, Theys N, Brenot H, Cayol V, Allard P, Caudron C, **Chevrel MO**, Darchambeau F *et al.* (2022) Precursor-free eruption triggered by edifice rupture at Nyiragongo volcano. *Nature 609*, 83-88. https://doi.org/10.1038/s41586-022-05047-8
34. Harris A, Mannini S, Calabrò L, Calvari S, Gurioli L, **Chevrel MO**, Favalli M, Villeneuve N. (2022) Forest destruction by 'a'a lava flow during Etna's 2002-03 eruption: Mechanical, thermal and environmental interactions. *Journal of Volcanology and Geothermal Research.* 429:107621. https://doi.org/10.1016/j.jvolgeores.2022.107621
33. Harris A, Rowland S., **Chevrel MO** (2022) Anatomy of a channel-fed 'a'a lava flow system. *Bulletin of Volcanology* 84:70 https://doi.org/10.1007/s00445-022-01578-0
32. Tadini A, Harris A, Morin J, Bevilacqua A, Peltier, Aspinall W, Ciolli, S, Bachelery P, Bernard B, Biren J, Brum da Silveiri A, Cayol V, **Chevrel MO**. Coppola D *et al.* (2022) Structured elicitation of expert judgement in real-time eruption scenarios: an exercise for Piton de la Fournaise volcano, La Réunion island. *Volcanica. 4(1):* 105 – 131. http://dx.doi.org/10.30909/vol.05.01.105131
31. Kolzenburg S, **Chevrel MO,** Dingwell D. (2022) Magma suspension rheology. *Reviews in Mineralogy and Geochemistry* 87 (1): 639–720. http://dx.doi.org/10.2138/rmg.2022.87.14
30. Ramírez-Uribe I, Siebe C, **Chevrel MO**, Ferrés D, Salinas S. (2022) The Late Holocene Nealtican lava-flow field from Popocatépetl volcano: Emplacement dynamics and implications for future hazard scenarios and archaeology. *GSA Bulletin. 134 (11-12): 2745–2766.* https://doi.org/10.1130/B36173.1
29. Reyes-Guzmán N, Siebe C, **Chevrel MO**, Pereira G, Mahgoub AN, Böhnel H. Holocene volcanic eruptions of the malpaís de zacapu and its pre-hispanic settlement history. *Ancient Mesoamerica. Accepted Oct. 2021*
28. Peltier A, **Chevrel MO**, Villeneuve N, Harris A (2022) Reappraisal of gap analysis for effusive crises at Piton de la Fournaise. *Journal of Applied Volcanology* 11, 2. https://doi.org/10.1186/s13617-021-00111-w
27. **Chevrel MO**, Favalli M, Villeneuve N, Harris A, Fornaciai A, Richter N. Derrien A, Boissier P, Di Muro A, Peltier A (2021) Lava flow hazard map of Piton de la Fournaise volcano. *Natural Hazards in Earth System Sciences,* 21, 1–22, 2021 https://doi.org/10.5194/nhess-21-2355-2021
26. Reyes-Guzmán N, Siebe C, **Chevrel MO**, Pereira P (2021) Late Holocene Malpaís de Zacapu (Michoacán, Mexico) andesitic lava flows: rheology and eruption properties based on LiDAR image. *Bulletin of Volcanology* 83, 28. https://doi.org/10.1007/s00445-021-01449-0
25. Ramírez-Uribe I, Siebe C, **Chevrel MO**, Fisher CT (2021) Rancho Seco monogenetic volcano (Michoacán, Mexico): Petrogenesis and lava flow emplacement based on LiDAR images – *Journal of Volcanology and Geothermal Research.* Vol 411:107169 https://doi.org/10.1016/j.jvolgeores.2020.107169
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MEDIA AND PUBLIC OUTREACH

2026 Conférence les Estivales de la Malepère
2026 Article The conversation
2026 Article dans La Montagne
2025 Article magasine La recherche
2025 Article Science & Vie, *Science & Vie n°1295*
2024 Mesurer la viscosité de la lave. Cycle des conference à Cité du volcan, La Réunion
2024 Etude des coulées de lave du Piton de la Fournaise. Séminaire formation volcan Enseignants du secondaire. Cité du volcan, La Réunion
2023 Organisation « Qué hace esta lava en mi reserve y de donde viene? » 40 anniversario de la Reserva Ecologica del Pedregal de San Angel, UNAM, Mexique
2023 Intervention « carrefour au féminin » collège Plateau Goyave, La Réunion
2023 Seminaire « Geosciences au féminin », Cité du volcan, La Réunion
2022,23,24 Fête de la science, Cité du volcan, *La Réunion*
2022 Public Q&A: Intervention Film Kraft CGA Les ambiances, *Clermont-Ferrand*
2022, 2021 Festival Nuées Ardentes, *Clermont-Ferrand*
2022 Public Q&A: "Rencontre Montagnes et Science", *Clermont-Ferrand*
2022 Seminar "Congres les femmes en Science : Des Volcans et des Femmes", *Paris*
2021 TV documentary Le monde de Jamy "Quand nos volcans se déchainent" *La Réunion*
2021 TV documentary RMC Story : "Les volcans d'Auvergne vont-ils se réveiller ?", *Clermont-Ferrand*
2021 Radio interview: L'infiniment Chaud. France Culture, Juillet 2021
2021 Youpi Magasine n°393 juin 2021
2020 Review "Sciences Animées Blog" de Christophe Wecker
2019 Intervention in High School "femmes ingénieures" *Lycée R Garros, La Réunion*
2019 Primary school workshop about being volcanologist, *Saint-Joseph, La Réunion*
2014 Primary school workshop about volcanoes, *Ecole Française, México*